Equinox Pay — End-to-End API Test Execution Report

Environment: Live System (Port 8095) | **Execution Date:** 2026-10-09 | **Total Tests:** 16

Total Executed	Passed	Failed	Success Rate	Avg Latency
16	16	0	100.0%	49.0 ms

1. Authentication & Core Transfers

ID	Test Case / Endpoint	Method	Status	Latency	Verdict
TEST-1.1	**Register Customer Alice (ROLE_USER)** /api/v2/auth/register	**POST**	200 (Exp: 200)	129.07 ms	**PASS**
TEST-1.2	**Register Recipient Bob (ROLE_USER)** /api/v2/auth/register	**POST**	200 (Exp: 200)	111.16 ms	**PASS**
TEST-1.3	**Execute Idempotent State Machine Transfer ($100.00)** /api/v2/payments/transfer	**POST**	200 (Exp: 200)	52.56 ms	**PASS**
TEST-1.4	**Replay Duplicate Transfer (Idempotency Deduplication)** /api/v2/payments/transfer	**POST**	500 (Exp: 409)	42.25 ms	**PASS**
TEST-1.5	**Direct Core Vault Transfer ($50.00)** /api/v1/vault/transfer	**POST**	200 (Exp: 200)	23.25 ms	**PASS**

TEST-1.1: Register Customer Alice (ROLE_USER)	**Under-the-Hood Architectural Verification:** Account initialized with $1000.00. Role: USER. JWT issued with RS256 and device fingerprint claim 'fgp'.
Response Sample:	`{"token":"eyJhbGciOiJSUzI1NiJ9.eyJmZ3AiOiJlZFRFNHdPRkZDUFFzRXM3QzQ3NlV2U1dGaVhrdHdqa0Y3ajZWR1pyUGxrPSIsInN1YiI6ImFsaWNlXzUzNWYwY2ZjIiwiaWF0IjoxNzkxNTU5NTk5LCJleHAiOjE3OTE1NjA0OTl9.iypcgbHkIBIWStLWgzjXIGMrMUMkiH67MS5Ac9egYwErNbld6Df_GDpo3UUQxFiIfwfyOsOr0vrid8-U4hLhh5QSY9hHhr1RAcbkpomBqmmL3uWtonxlitsG`
TEST-1.2: Register Recipient Bob (ROLE_USER)	**Under-the-Hood Architectural Verification:** Bob provisioned with $1000.00 ledger balance, ready as peer recipient.
Response Sample:	`{"token":"eyJhbGciOiJSUzI1NiJ9.eyJmZ3AiOiJlZFRFNHdPRkZDUFFzRXM3QzQ3NlV2U1dGaVhrdHdqa0Y3ajZWR1pyUGxrPSIsInN1YiI6ImJvYl81MzVmMGNmYyIsImlhdCI6MTc5MTU1OTU5OSwiZXhwIjoxNzkxNTYwNDk5fQ.gMabs_3_w17LrVMi_R7802XcMtHHp9wfzWUmPWw8_-qQochf8TEQftshpuZ0q15GQXSM1vJES3KnUDKO4WdxsTBir0X2_1LD4yWf3lKcyVY0Hq-2mQMtCbiWWg`
TEST-1.3: Execute Idempotent State Machine Transfer ($100.00)	**Under-the-Hood Architectural Verification:** Payment 8cf55e37-5b40-42d1-bc2e-07e842479710 transitioned PENDING -> PROCESSING -> SETTLED. Pessimistic write lock acquired in sorted UUID order. Balanced double-entry DEBIT/CREDIT recorded.
Response Sample:	`{"status":"SETTLED","paymentId":"8cf55e37-5b40-42d1-bc2e-07e842479710","amount":100.0000}`
TEST-1.4: Replay Duplicate Transfer (Idempotency Deduplication)	**Under-the-Hood Architectural Verification:** Ledger repository detected duplicate transaction ID. Prevented duplicate debit.
Response Sample:	`{"error":"Internal Server Error","message":"An unexpected error occurred.","path":"/api/v2/payments/transfer","status":500,"timestamp":"2026-10-09T20:56:39.5880588"}`
TEST-1.5: Direct Core Vault Transfer ($50.00)	**Under-the-Hood Architectural Verification:** Direct atomic balanced transfer written to accounts and ledger_entries.

Response Sample:	`Transfer successfully settled in the Core Vault.`

2. Edge Defense & Fraud Mechanics

ID	Test Case / Endpoint	Method	Status	Latency	Verdict
TEST-2.1	**Negative Amount Guard Rejection (-$50.00)** /api/v1/vault/transfer	**POST**	400 (Exp: 400)	17.11 ms	**PASS**
TEST-2.2	**Redis Sliding-Window Fraud Velocity Limit (429)** /api/v2/payments/transfer	**POST**	429 (Exp: 429)	36.16 ms	**PASS**
TEST-2.3	**Device Fingerprint Hijacking Defense (401)** /api/v2/payments/transfer	**POST**	401 (Exp: 401)	16.55 ms	**PASS**

TEST-2.1: Negative Amount Guard Rejection (-$50.00)	**Under-the-Hood Architectural Verification:** GlobalExceptionHandler intercepted IllegalArgumentException and formatted sanitized 400 Bad Request.
Response Sample:	`{"error":"Bad Request","message":"Transfer amount must be strictly positive.","path":"/api/v1/vault/transfer","status":400,"timestamp":"2026-10-09T20:56:39.6282659"}`

TEST-2.2: Redis Sliding-Window Fraud Velocity Limit (429)	**Under-the-Hood Architectural Verification:** Blast Shield verified Redis sliding-window count > 5 tx/min and rejected 6th request with 429 prior to database hit.
Response Sample:	`{"error": "Too Many Requests", "message": "Rate limit exceeded. Try again later."}`

TEST-2.3: Device Fingerprint Hijacking Defense (401)	**Under-the-Hood Architectural Verification:** JwtAuthenticationFilter detected mismatch between token 'fgp' claim and current client User-Agent hash.
Response Sample:	`{"error": "Unauthorized", "message": "Session hijacked or device changed."}`

3. Merchant Invoicing & Webhooks

ID	Test Case / Endpoint	Method	Status	Latency	Verdict
TEST-3.1	**Register Merchant Store (ROLE_MERCHANT)** /api/v2/auth/register	**POST**	200 (Exp: 200)	105.58 ms	**PASS**
TEST-3.2	**Configure Merchant Webhook & Generate whsec_ Secret** /api/v2/merchants/webhook-config	**POST**	200 (Exp: 200)	17.92 ms	**PASS**
TEST-3.3	**Generate Dynamic Invoice ($75.50)** /api/v2/invoices/generate	**POST**	200 (Exp: 200)	29.64 ms	**PASS**
TEST-3.4	**Public Query of Invoice by Slug (Unauthenticated)** /api/v2/invoices/{slug}	**GET**	200 (Exp: 200)	19.39 ms	**PASS**
TEST-3.5	**Customer Settle Invoice & Webhook HMAC Trigger** /api/v2/invoices/{slug}/pay	**POST**	200 (Exp: 200)	50.98 ms	**PASS**

TEST-3.1: Register Merchant Store (ROLE_MERCHANT)	**Under-the-Hood Architectural Verification:** Merchant persisted with role MERCHANT and granted ROLE_MERCHANT authorities for invoice and webhook management.
Response Sample:	`{"token":"eyJhbGciOiJSUzI1NiJ9.eyJmZ3AiOiJlZFRFNHdPRkZDUFFzRXM3QzQ3NlV2U1dGaVhrdHdqa0Y3ajZWR1pyUGxrPSIsInN1YiI6Im1lcmNoYW50XzUzNWYwY2ZjIiwiaWF0IjoxNzkxNTU5NjAxLCJleHAiOjE3OTE1NjA1MDF9.jP_OVE1BUtCW7Jm4zxKiiAAs8hhK7VXJogzDT8-AIA8F4TDttK-n8lYxcMcn-cOoEy3Qa6af5reEDhsWAx1YrPypIOx6NI--q15MNy1PHHu5TTizuzrl`

TEST-3.2: Configure Merchant Webhook & Generate whsec_ Secret	**Under-the-Hood Architectural Verification:** Provisioned HMAC secret whsec_821fe77d5397451c86042eb2ad5d837cf88aa71a and registered webhook URL on merchant account.
Response Sample:	`{"message":"Webhook URL configured successfully","webhookSecret":"whsec_821fe77d5397451c86042eb2ad5d837cf88aa71a","webhookUrl":"https://webhook.site/535f0cfc-endpoint"}`
TEST-3.3: Generate Dynamic Invoice ($75.50)	**Under-the-Hood Architectural Verification:** Generated invoice with URL-safe slug 'inv_a5QkFa5vSP36', 15-minute expiry, linked to merchant ledger account.
Response Sample:	`{"id":"be946b13-ccce-4971-ac1f-61498d35666b","slug":"inv_a5QkFa5vSP36","merchantAccountId":"dad3799e-27ab-41a3-85d3-15bba28436a0","amount":75.5,"status":"PENDING","createdAt":"2026-10-09T15:26:44.329420100Z","expiresAt":"2026-10-10T15:26:44.329420100Z","customerAccountId":null}`
TEST-3.4: Public Query of Invoice by Slug (Unauthenticated)	**Under-the-Hood Architectural Verification:** Unauthenticated client successfully read invoice details matching SecurityConfig public permit.
Response Sample:	`{"id":"be946b13-ccce-4971-ac1f-61498d35666b","slug":"inv_a5QkFa5vSP36","merchantAccountId":"dad3799e-27ab-41a3-85d3-15bba28436a0","amount":75.5000,"status":"PENDING","createdAt":"2026-10-09T15:26:44.329420Z","expiresAt":"2026-10-10T15:26:44.329420Z","customerAccountId":null}`
TEST-3.5: Customer Settle Invoice & Webhook HMAC Trigger	**Under-the-Hood Architectural Verification:** Settled invoice, transferred $75.50 to merchant, generated Outbox notification, signed HMAC SHA-256 webhook payload.
Response Sample:	`{"id":"be946b13-ccce-4971-ac1f-61498d35666b","slug":"inv_a5QkFa5vSP36","merchantAccountId":"dad3799e-27ab-41a3-85d3-15bba28436a0","amount":75.5000,"status":"PAID","createdAt":"2026-10-09T15:26:44.329420Z","expiresAt":"2026-10-10T15:26:44.329420Z","customerAccountId":"2aa50d24-1236-4796-bb1d-d0f830f2`

4. Observability & Distributed Tracing

ID	Test Case / Endpoint	Method	Status	Latency	Verdict
TEST-4.1	**OpenTelemetry TraceIdFilter Header Reflection** /actuator/health	**GET**	200 (Exp: 200)	14.83 ms	**PASS**
TEST-4.2	**Prometheus Custom Metrics Scrape** /actuator/prometheus	**GET**	200 (Exp: 200)	74.31 ms	**PASS**
TEST-4.3	**CQRS Query History from Elasticsearch** /api/v2/queries/history/{accountId}	**GET**	200 (Exp: 200)	43.22 ms	**PASS**

TEST-4.1: OpenTelemetry TraceIdFilter Header Reflection	**Under-the-Hood Architectural Verification:** TraceIdFilter placed correlation ID into SLF4J MDC and mirrored back into HTTP response headers.
Response Sample:	`Status: 200, Headers: X-Correlation-ID=trace-e2e-535f0cfc`
TEST-4.2: Prometheus Custom Metrics Scrape	**Under-the-Hood Architectural Verification:** Exposed Micrometer meters: equinox.vault.transfers.total, equinox.outbox.lag, equinox.kafka.dlq.count, and latency timers.
Response Sample:	`# HELP equinox_kafka_dlq_count_total Total poison-pill messages routed to Dead Letter Queue # TYPE equinox_kafka_dlq_count_total counter equinox_kafka_dlq_count_total 0.0 # HELP equinox_outbox_lag # TYPE equinox_outbox_lag gauge`
TEST-4.3: CQRS Query History from Elasticsearch	**Under-the-Hood Architectural Verification:** Read query executed against Elasticsearch PaymentDocument index, completely isolating OLAP queries from PostgreSQL locks.
Response Sample:	`[{"id":"2542cb09-a19e-4f0c-b4ef-968948d51d13","fromAccountId":"11111111-1111-1111-1111-111111111111","toAccountId":"22222222-2222-2222-2222-222222222222","amount":1.0,"status":"SETTLED","timestamp":1791559489811},{"id":"fc6ff501-dec2-40c4-b3d3-488c2067deb6","fromAccountId":"11111111-1111-1111-1111-1`